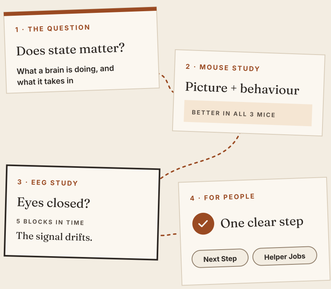
What this is

Does what a brain is doing change what it takes in?

Two studies with public brain recordings: a mouse watching pictures, and a person opening and closing their eyes. Each result is tested on data kept aside, then checked against a control that could break it.



The mouse study → · The EEG study

Public data only. Prediction is not cause. Plans are marked as plans.

- **Mouse visual cortex** — Pupil and running help predict it
- **EEG eye state** — Good once, then it drifts
- **Every result checked** — Shuffles, time gaps, one left out
- **Tools for people** — Next Step and Helper Jobs

Five next experiments, each with the same steps.

Two studies, two tools

Each starts with a question we could be wrong about, and ends with something a person can read, check or use.

STUDY 1 · DONE

Does knowing what a mouse is doing help predict its visual cortex?

A model that sees only the picture, against one that also knows pupil size and running speed. Three mice, one held-out test.

Result: better in all three mice, by +0.032 to +0.049. Open the study →

STUDY 2 · DONE

Can brain waves tell when eyes are closed?

One person, 14 EEG channels, 117 seconds. Always trained on the past and tested on the future.

Result: 0.625 on the last block (0.728 without four glitches), but near a coin flip across time. Open the study →

TOOL · BY AMISHI MEHTA

Next Step

Long instructions, interruptions and explaining your needs are hard for many neurodivergent young people.

Next Step: one next step, words to ask for help, and a note of where to restart. See the tools →

TOOL · FAMILY PROJECT

Helper Jobs

At Pragnya's Sunday outings, "help out" is not a job, and spoken instructions are hard to hold on to.

Helper Jobs: a real job in picture steps, with a volunteer close by. See the tools →

Under all of it, honesty: results are kept even when they disappoint, and every number on these pages is generated from committed results. Three mice and one EEG recording cannot speak for all mice or all people.

What this can't tell us yet: whether these patterns hold in more animals and more people, or what causes them. Next: five experiments with the same steps.



How we tried to break each result

A result is only shared after a check that could have made it disappear.

SHUFFLE THE ANSWERS

A coin flip, as it should be

With the EEG labels scrambled 20 times, the score falls to 0.499. Mouse behaviour shuffled 20 times adds almost nothing.

SPLIT BY TIME

Train on the past, test the future

No random mixing of neighbouring moments, which makes EEG results look far better than they are.

LEAVE A GAP

90 samples between train and test

Removing the moments right next to each test block keeps the EEG result near chance: it is not leakage.

LEAVE ONE OUT

No single mouse carries it

The average gain stays positive (+0.040 to +0.048) whichever mouse is left out.

GLITCH CHECK

Four broken samples found

Values up to 715,897 where normal is about 4,000. Limiting them helps the last block, not the drift.

EVERY NUMBER TRACED

Rebuilt from saved results

The charts are generated from the committed results files, and the EEG study reruns with one command.

The results, at a glance

Three results from the studies. Open a study for the charts and the checks.



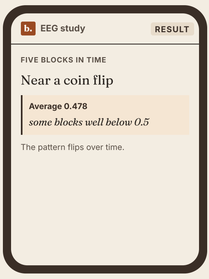
Better in every mouse

Adding pupil and running raised prediction by about a quarter to a third.



Good on one block

The simple model read eye state well on the last part of the recording.



But it drifts

Over the 117 seconds the link between signal and eye state changes.

Who did what

The studies were done by the applicant with a parent collaborator, who helped scope questions, review code and debug. Software tools, including AI assistants, helped build this website. The contribution ledger lists each part.

What it is not

It is not a clinical or diagnostic study. Three mice and one EEG recording cannot speak for all animals or people, and a model that predicts something does not show what causes it.

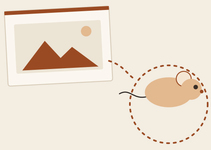
Data: Sensorium 2022 (Willeke et al., 2022) and UCI EEG Eye State (Roesler, 2013). Mouse study · EEG study · Tools · Next experiments

← All studies

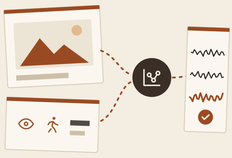
STUDY 1 · MOUSE VISUAL CORTEX · DONE

Does knowing what a mouse is doing help predict what its brain sees?

A model that sees only the picture, against one that also knows the mouse's pupil size and running speed.



The answer: yes, in all three mice. Adding pupil and running raised held-out prediction by +0.032 to +0.049, about 25–34% more than the picture alone.
This is prediction, not proof of cause.



How we tested it

The same steps for every mouse, scored on images the model never saw.

THE PICTURE

Shrunk and summarised

Each image is shrunk 4×4 and summarised with 150 principal components.

THE BEHAVIOUR

Pupil and running

Pupil size and running speed recorded at the same moment as the brain activity.

THE MODEL

Ridge regression

One simple model per neuron, standardised on training data only.

THE TEST

Official held-out images

Scored by the correlation between predicted and real activity, averaged over neurons.

THE DATA

Sensorium 2022

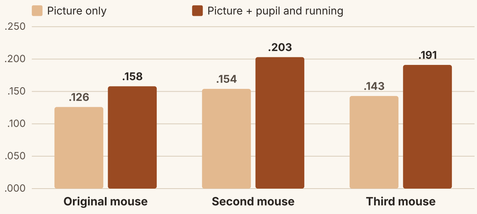
Public recordings from mouse primary visual cortex, thousands of neurons per mouse.

THE UNIT

Each mouse counts once

Neurons are not treated as separate animals.

The results

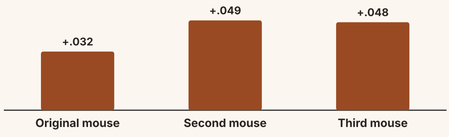


RESULT

Every mouse is predicted better with behaviour.

The picture alone explains part of what visual cortex does. Knowing whether the mouse is alert and running explains more, in all three animals.

Mean held-out Pearson correlation across neurons, Sensorium 2022 official test images.



GAIN

The size of the gain differs by mouse.

Pupil size helps all three animals. How much running adds changes from mouse to mouse, so each animal is reported on its own.

Gain = score with behaviour minus score with the picture only.

Trying to break the result

Three checks that could have made the gain disappear.



SHUFFLE THE BEHAVIOUR

Almost no gain

Behaviour scrambled across trials, 20 times per mouse, adds almost nothing. The gain needs behaviour from the same moment.

LEAVE ONE MOUSE OUT

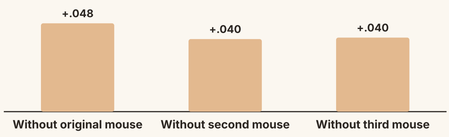
Still positive

The average gain stays positive whichever mouse is removed, so no single recording carries the result.

PUPIL OR RUNNING?

Pupil helps every mouse

Taking each signal away in turn: pupil improves all three mice; running varies by animal.



LEAVE ONE MOUSE OUT

Positive without any one animal.

This makes the three-mouse pattern less dependent on one recording. It is a check, not an estimate for all mice.

Average gain over the two remaining mice.

Code and data

The analysis is in Amishl123/neuroscience-project (notebooks 01–05 and `src/summarize_replication.py`). The numbers here are copied from its committed summary into `website/build_results.py`. Data: Willeke et al. (2022), *The Sensorium competition on predicting large-scale mouse primary visual cortex activity*, arXiv:2206.08666.

What it does not show

Three mice cannot speak for all mice. A better prediction does not show that arousal or running cause the change. The report's intervals are within each animal, not for the whole population.

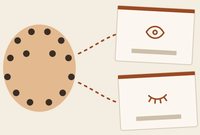
← All studies EEG study →

← All studies

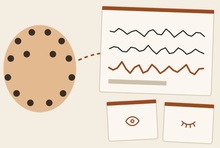
STUDY 2 · EEG EYE STATE · DONE

Can brain waves tell when your eyes are closed?

And does giving the model the last few moments of signal help it decide?



The answer: sometimes, but not reliably over time. The model scored 0.625 on the last part of the recording, 0.728 once four glitch samples were limited, yet across five blocks in time it was close to a coin flip.
Recent history did not help.



How we tested it

Always train on the past, and test on the future.

THE RECORDING

One person, 117 seconds

14 channels and 14,980 samples, about 45% with eyes closed.

SPLIT BY TIME

70 · 15 · 15

Learn on the first 70%, choose on the next 15%, test once on the last 15%. No random mixing.

TWO MODELS

A moment, or a moment plus history

One sees a single moment; the other also sees the average and spread of the last 9 samples.

THE SCORE

ROC AUC

0.5 is a coin flip and 1.0 is perfect.

FIVE BLOCKS

Forward in time

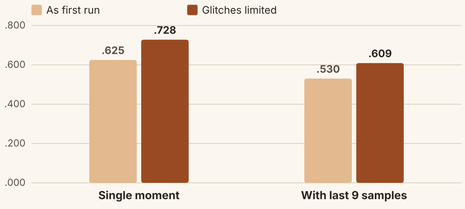
Train on everything before a block, test on the block, five times.

THE DATA

UCI EEG Eye State

Roesler (2013), a public recording with each moment marked eyes open or closed.

The results

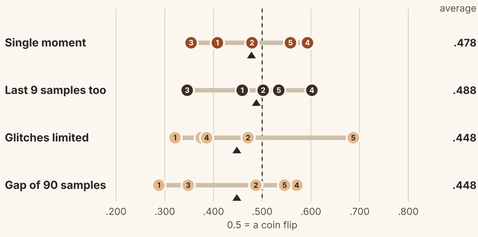


FINAL TEST BLOCK

Good on one block, better without glitches.

Four samples are recording glitches, with values up to 715,897 where normal readings are about 4,000. Limiting each channel to the range seen in training lifts the simple model from 0.625 to 0.728.

The glitch check was added after the first results, so it is shown next to the original numbers, not instead of them.



FIVE BLOCKS IN TIME

Over time, it drifts to a coin flip.

Scores scatter around 0.5, and several blocks fall well below it. A model that is reliably worse than chance has learned a pattern that flips later in the recording.

That points to drift across the 117 seconds, not only noise. Limiting glitches and leaving a time gap do not change it.



Making sure it is not the pipeline

Three checks on how the result was produced.

SHUFFLE THE ANSWERS

0.499 average

Eye-state labels scrambled 20 times give a coin flip, as they should. The pipeline does not invent signal.

LEAVE A GAP

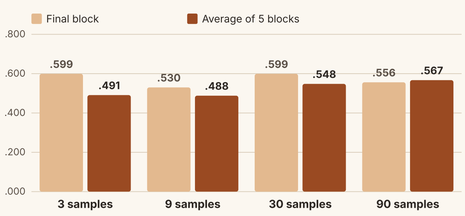
0.448 average

Removing the 90 samples before each test block keeps the result near chance, so it is not leakage from neighbours.

LIMIT THE GLITCHES

Rows 898, 10386, 11509, 13179

Limits are learned from training rows only, separately for each block, so the test never shapes them.



HOW MUCH HISTORY

Longer history looks better, but we did not pick a winner.

The 3, 9, 30 and 90-sample windows were set before looking. Ninety does best on average, but choosing it now would be fitting this one recording.

Next: test window lengths on new recordings before choosing.

Rerun it

`python3 projects/eeg-eye-state/src/analyze.py` writes `results/metrics.json`, and `python3 website/build_results.py` turns it into these charts. See the full report. Data: Roesler, O. (2013). *EEG Eye State*. UCI Machine Learning Repository, doi:10.24432/C57G7J.

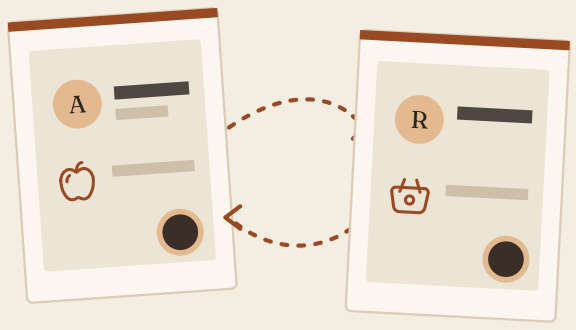
What it does not show

One person and one short recording cannot show how this works for other people. The label is eyes open or closed, not attention, alertness or any condition.

← All studies

APPLIED WORK · TOOLS FOR PEOPLE

Turning what we learn into one clear step at a time.



Attention and state shape what a brain takes in. These tools keep each step small and calm, and nothing happens until the person taps.

Two working apps

Both use the same design: big type, picture steps, read aloud, and made-up examples only. Swipe to see both.

BY AMISHI MEHTA

Next Step

For anyone who finds long instructions, interruptions or explaining their needs hard. Made with neurodivergent young people in mind.

What it does: turns a long brief into one next step, practises words for asking for help, and keeps a note of where to restart. Open Next Step →

FAMILY PROJECT FOR PRAGNYA

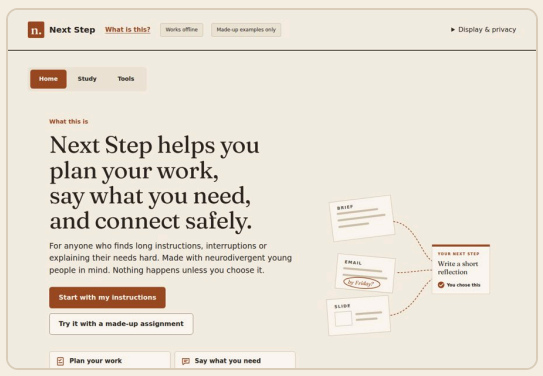
Helper Jobs

Gives neurodivergent adults a real job at kids' Sunday outings, with a volunteer always close by.

What it does: picture steps read aloud in English, Hindi or Spanish, and Help that reaches the lead's phone in about a second. Open Helper Jobs →

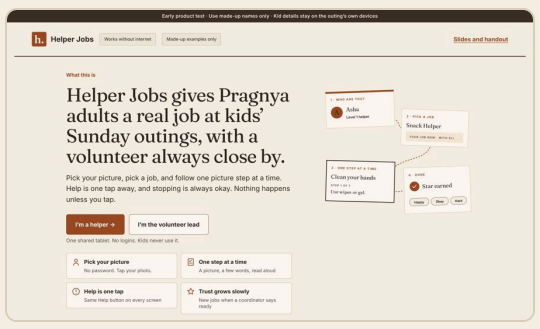
The apps

Real screens from both apps.



Next Step

Plan your work, say what you need, and connect safely.



Helper Jobs

Pick your picture, pick a job, one step at a time.



One picture step

A few words, a picture of the real place, and Read aloud.

Research demo on a phone

The mobile research demo puts both studies in a pocket-sized tour.

Careful claims

These are early test versions with made-up people in every example. They have not been tested in a formal study, and they do not diagnose or treat anything. A usability study is planned as experiment 5.

← All studies

WHAT COMES NEXT

Five experiments, each from question to explanation.

Each gets an analysis written down before it runs, a held-out test, a control that could break it, and an honest write-up.



The five experiments

1 · PLANNED

Mouse behaviour timing

Does behaviour from the same trial predict activity better than behaviour shifted to neighbouring trials? Needs the raw Sensorium recordings, which are not in this repository.

2 · READY TO RUN

EEG frequency bands

Do brain-wave bands, like alpha that rises when eyes close, predict eye state better than raw signals over time? Uses the EEG file already here.

3 · PLANNED

Mouse neuron groups

Is the gain spread across neurons, or concentrated in groups chosen in advance by depth and position? Needs raw responses and neuron positions.

4 · STARTED

EEG drift over time

How stable is prediction across the recording, and does adjusting for drift help? Block-by-block scores and the glitch check are done.

5 · PLANNED

Usability with people

Do the step-by-step designs make these research pages easier to follow? Needs consent, a task list written in advance and no unneeded personal data.

The rule we keep: a plan stays labelled as a plan until it has code, saved results, a rerun record, its limits written down and a page like the two studies.

← Tools for people All studies →